Using the formula for the area of a circle

Task A: Finding areas of circles by using the formula

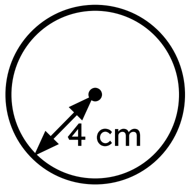
1) Find the area of each circle, giving your answers as decimals rounded to 3 significant figures. Diagrams are not drawn to scale.

a) circle with radius 6 m

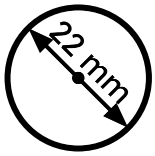
area = 113 m^2

b) circle with diameter 6 m

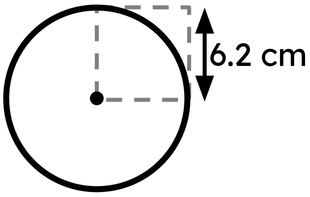
area = 28.3 m^2

c) 

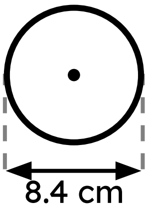
area = 50.3 cm^2

d) 

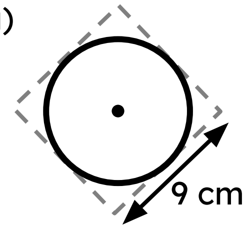
area = 380 mm^2

e) 

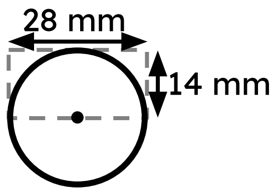
area = 121 cm^2

f) 

area = 55.4 cm^2

g) 

area = 63.6 cm^2

h) 

area = 616 mm^2

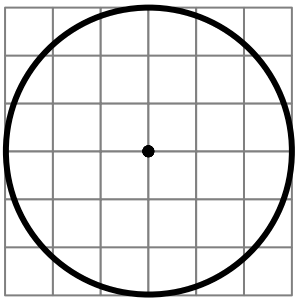
2) Find the area of each circle, giving your answers in terms of π.

a) The distance from the centre of the circle to the circumference is 12 cm.

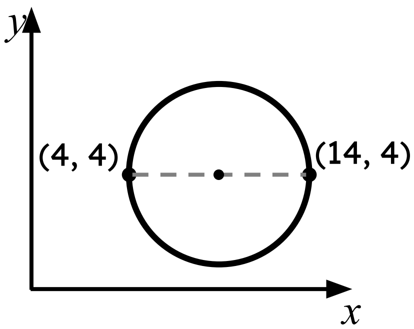
area = *144π* cm^2

b) The furthest distance from one side of the circle to the other is 12 cm.

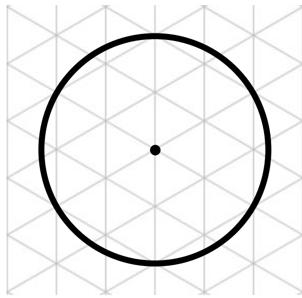
area = *36π* cm^2

c) 

area = *9π* units^2

d) 

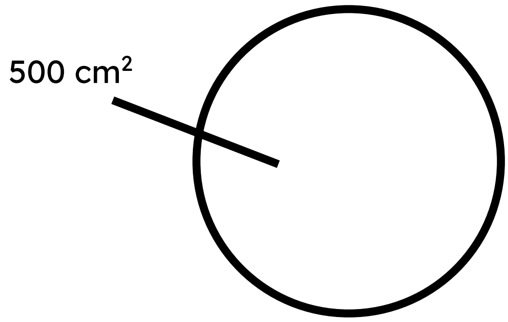
area = *25π* units^2

e) 

area = *4π* units^2

Task B: Finding the radius or diameter when the area is known

1) The area of the circle is 500 cm^2.
For each question, give your answer to 1 decimal place.



a) Calculate the radius. 12.6 cm

b) Calculate the diameter. 25.2 cm

c) Calculate circumference. 79.3 cm